

Hamna Saleem

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PROFESSIONAL SUMMARY & CAREER GOAL

Final-year Biomedical Engineering (Honours) student at UTS and Biomedical Electronics Technician at Royal Prince Alfred Hospital, with experience across clinical engineering, biotechnology research and medical-device development. My experience across clinical, research and design environments allows me to understand healthcare problems from multiple perspectives. I bring this perspective to medical-device teams to identify real clinician and patient needs and translate them into practical, technically informed solutions. I aim to build my career in medical-device R&D, developing technologies that are effective, practical and accessible in real healthcare settings.

EDUCATION

Bachelor of Engineering (Honours)
Diploma in Professional Engineering Practice
University of Technology Sydney

Feb 2022 – Dec 2026

Major in Biomedical

- Current GPA 6.82 (High Distinction Average)
- Dean's Merit List 2025
- High Distinction in Electronics and Circuits
- Distinction in Medical Devices and Diagnostics
- High Distinction in Biomedical Instrumentation

WORK EXPERIENCE

Biomedical Electronics Technician

Jul 2026 – Present

- Perform preventative maintenance, testing, verification and fault diagnosis of medical equipment in accordance with AS/NZS 3551, supporting equipment safety and reliability.
- Respond to equipment faults and clinician concerns across clinical areas, including the Emergency Department, troubleshooting issues and coordinating technical support.
- Manage medical equipment assets and servicing, coordinating with clinical teams and external providers to maintain equipment availability.

Biomedical Engineering Intern

Jan 2026 – May 2026

EnGeneIC

- Produced bispecific antibodies for targeted cancer therapy through CHO cell culture, ÄKTA purification, sterile filtration and quality control.
- Quantified and assessed purified antibodies to support downstream research and evaluation.
- Troubleshoot experimental results and adapted laboratory methods using evidence-based decision-making.

Biomedical Engineering Intern

Jan 2024 – Jul 2024

Royal Prince Alfred and Concord Hospital, Sydney Local Health District

- Repaired, maintained and verified medical equipment in accordance with AS/NZS 3551 to support safe clinical use.
- Performed preventative maintenance and equipment audits across clinical areas, coordinating with hospital staff to minimise disruption.
- Coordinated with vendors and contractors to support medical-device servicing and procurement.

EMPLOYMENT

Operations Team Member

Sep 2021 - Present

Bunnings Warehouse, Bankstown Airport

- Lead and coordinate a team of 6–8 employees in a dynamic environment, ensuring operational efficiency and staff well-being.
- Manage customer enquiries through collaboration with internal teams to deliver clear, solution-focused communication.
- Adapt to changing demands and resolve customer enquiries using product knowledge and practical problem-solving to deliver high-quality customer service.

EXTRACURRICULAR ACTIVITIES

Undergraduate Student Representative

Mar 2025 – Jan 2026

University of Technology Sydney, Faculty of Engineering and IT

- Collected and represented student feedback to improve inclusivity and engagement across classes and faculty activities.
- Liaised with staff and student stakeholders to communicate issues and support student-informed improvements.

Mentee

Feb 2025 – Dec 2025

UTS Lucy Mentoring Program

- Participated in industry mentoring, professional-development workshops and mentor-guided activities focused on engineering career development.

PROJECTS

StrikeDX – LFA for Rapid Differentiation of Ischemic and Haemorrhagic Strokes

2025

Team Leader

- Developed a rapid point-of-care diagnostic using LFA to differentiate ischemic and haemorrhagic stroke within 15 minutes.
- Integrated dual biomarkers (GFAP & NT-proBNP) for improved accuracy and designed a user-friendly emergency-use cassette.
- Applied biomarker selection, immunoassay and fluorescence-detection principles to support diagnostic development.

Maternix – Portable PCR Unit for Remote Field Testing of Group B Streptococcus

2025

Research and Development Team Member

- Designed a portable point-of-care PCR diagnostic for Group B Streptococcus screening in remote and low-resource settings.
- Defined system requirements and architecture incorporating requirements traceability, risk-aware design and verification planning.
- Developed ESP32-based thermal control, sensing and Bluetooth communication for device operation and live monitoring.

Wireless Pulse Oximetry and Temperature Telemetry System

2025

Research & Development Member

- Built an embedded sensing prototype to measure PPG-based pulse metrics and temperature, with wireless data transmission for live monitoring.
- Implemented signal conditioning and beat-detection logic to improve measurement reliability, then benchmarked outputs against a commercial reference device.

TECHNICAL SKILLS

- **Electrical & Instrumentation:** Circuit troubleshooting, sensors, microcontroller-based systems, signal conditioning, verification, soldering
- **Biomedical Engineering:** Medical device servicing, equipment compliance (AS/NZS 3551)
- **Programming & Data Analysis:** Java, KNIME, MATLAB

- **Design & Simulation:** SolidWorks, LabVIEW, 3D printing
- **Image Processing & Analysis:** ImageJ
- **Software & Documentation:** MS Office, Google Suite

REFERENCES

Available on request.